

How AI/ML can Automate DevOps

AI/ML integration in DevOps can be transformative for software development. It can enable faster code development, improve the efficiency and reliability of the software, and automate the analysis of massive amounts of data.



Areas of application of AI/ML in DevOps

Some notable areas of application of AI/ML in DevOps are:

- Faster code development
 - Tools like GitHub Copilot and AWS Code Whisperer can help expedite code development
 - Identification of irregular code patterns
- Integrated unit test development
 - Creation of automated unit test cases using AI/ML tools
 - Tools like PyPenguin can be used to generate unit test cases for Python
- Code debugging
 - Automatic detection of bugs in code
- Code reviewing
 - Automated code reviews to identify potential security vulnerabilities, memory leaks, etc
- Robotic process automation
 - Creation of codeless automation macros for regression testing
- Continuous monitoring
 - Identification of behavioural patterns and irregularities
 - Differentiation between normal and abnormal patterns
 - Identification and prioritisation of alerts
 - Identification of hidden patterns and trends
- Anomaly detection
 - Detection of outliers
 - Normalisation of unexpected alerts
 - Detection of DDoS attacks
- Predictive analytics
 - Early actions based on predictive insights

The concept of DevOps was born with the purpose of fostering collaboration between the development and operations teams in order to streamline the software development life cycle. Numerous organisations are already reaping the benefits of DevOps. And now, AI and ML are being increasingly used in DevOps pipelines. The aim is to increase the operational efficiency and productivity. A shift towards automation can improve the overall performance of the system. In order to get optimum results, engineering teams adopt AI and ML techniques that can make maximum use of the underlying data and improve the overall DevOps experience.

Steps to include AI/ML in DevOps pipelines

By integrating AI/ML in DevOps,

massive amounts of data generated in the DevOps pipeline can be analysed. Different AI/ML algorithms that suit the needs of the organisation can be applied. Here are the key steps:

- Define the goal to use AI/ML in DevOps
- Identify the relevant tools and technologies
- Leverage cloud providers' APIs if available
- Analyse DevOps data and gather appropriate training and testing data
- Correlate different sets of inputs to find the best course of action
- Build and finalise the AI/ML models
- Train the ML models with the DevOps data to match with actuals
- Test model with the test data
- Optimise the model to cater to the dynamic needs of the operations